

Lab Report 3 - Cybersecurity Analytics

Analyzing a Packet to Assist in an Investigation

Introduction

This lab is designed to work with VM machines to get practical skills by analyzing a packet to assist in an investigation using Wireshark and Network Miner. Today most of the business, institutions and almost all the companies deal with cyber-attacks, which aim to disrupt services and compromise the availability of the systems. The XFibility bank also experienced same kind of incident. The bank customers are enabled to access their accounts through the bank web portal. The firewall at the bank shows a massive spike in traffic to the web server. The security operations center (SOC) at XFibility Bank declared a cybersecurity incident (Minnesota State University Moorhead, 2025). To investigate the disruption, a network traffic capture *file_lab2.pcap* was obtained.

Tools like Wireshark and Network minor can investigate to identify indicators of compromise. They can give details about victims IP addresses, HTTP GET requests, and the IP addresses which tried to disrupt the network. We can compare the observed traffic patterns with security control framework's best practices. NIST SP 800-53 Revision 5 (National Institute of Standards and Technology [NIST], 2020) describes how to reconstruct attack's progression and recommendation to improve defenses. Analysis is a dynamic process, methods use to analyze each scenario may not be the same (Sanders, 2017).

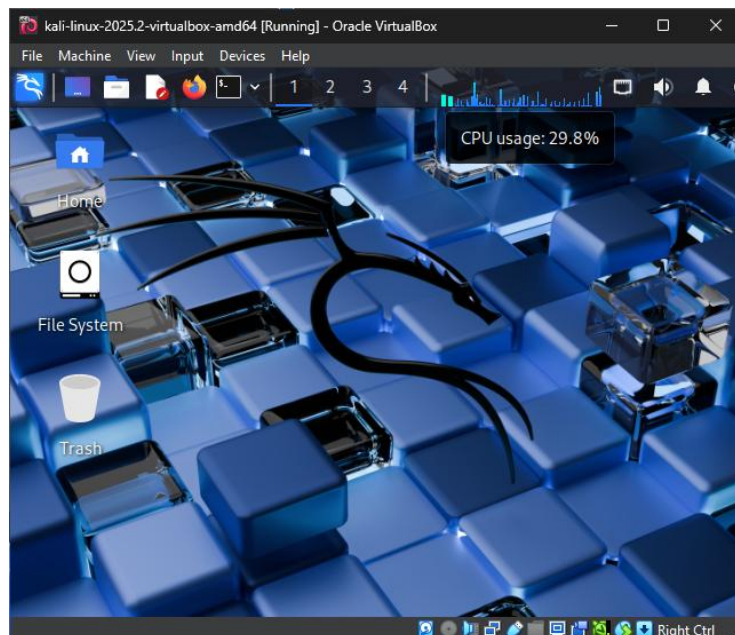
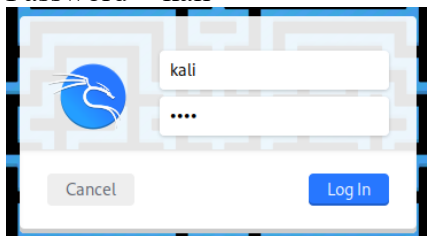
Step - 1

1. Logged to Azure Lab environment using following web link and logged into the Kali VM using Oracle VirtualBox.

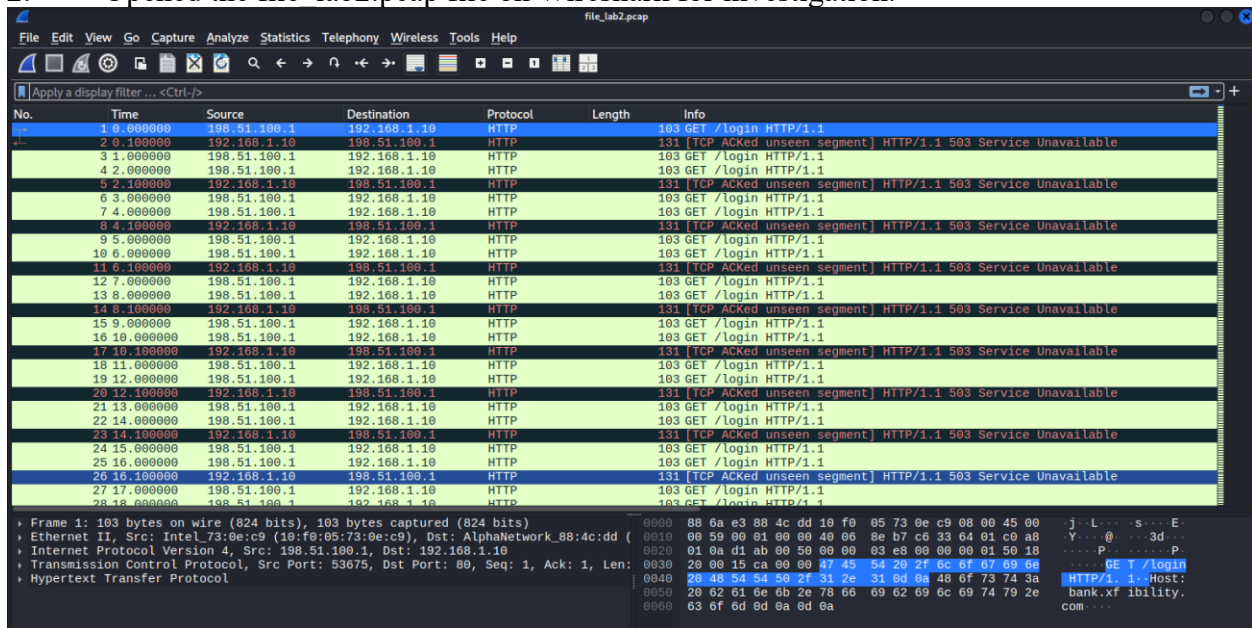
https://labs.azure.com/virtualmachines?feature_vnext=true

Username - kali

Password - kali



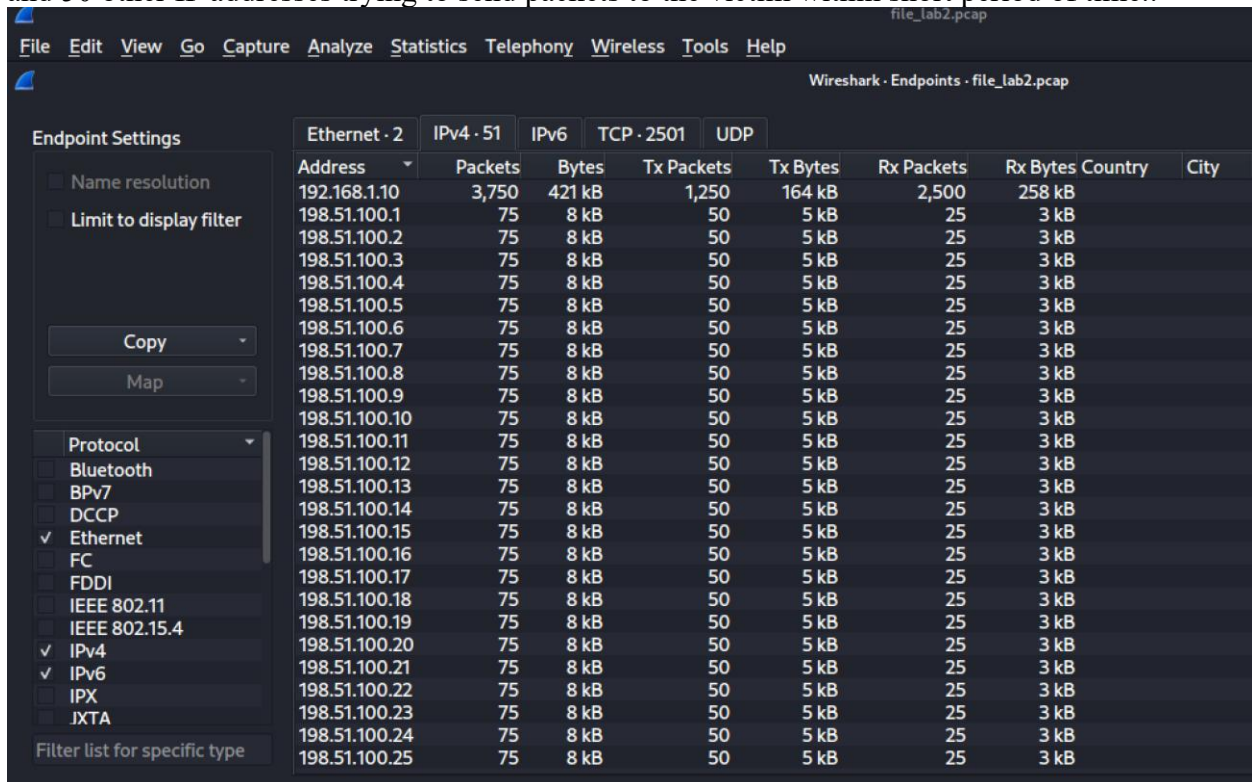
2. Opened the file lab2.pcap file on Wireshark for investigation.



3. What are the indicators of compromise you see in the pcap?

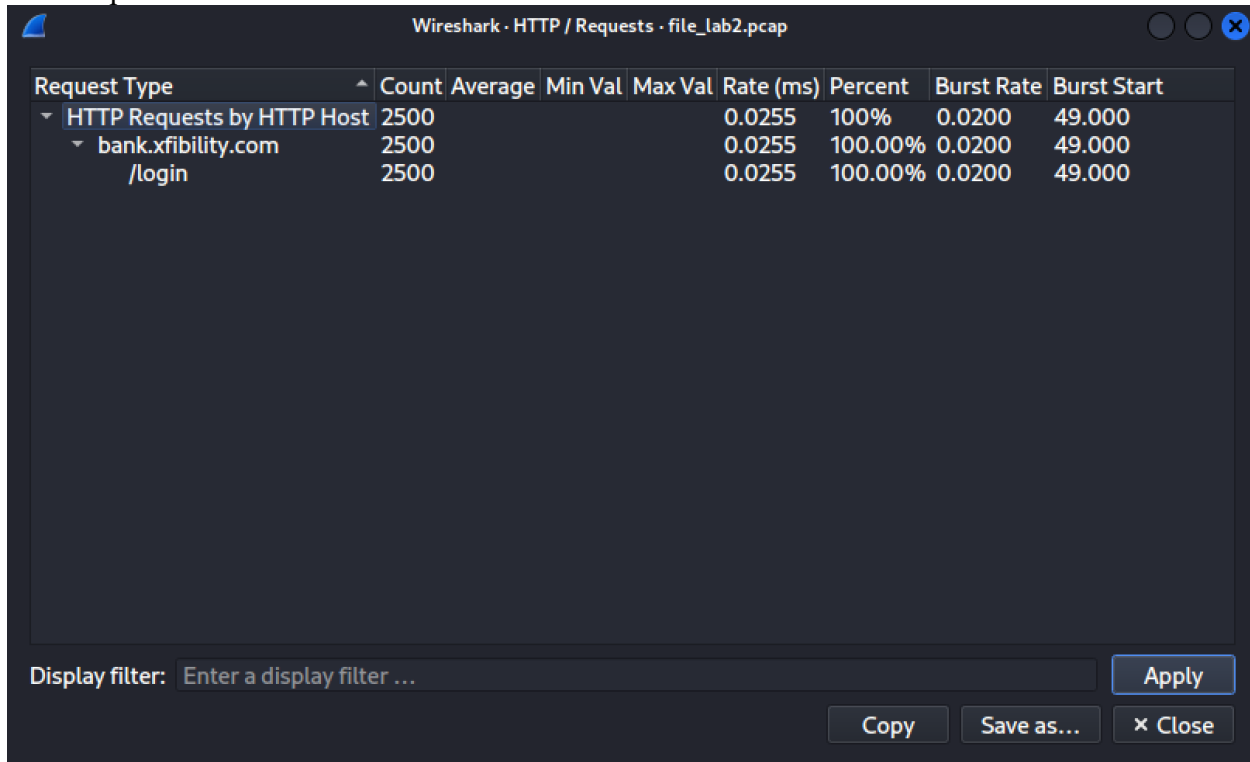
- From above capture we can see a lot of HTTP GET requests from multiple sources, and HTTP/1.1 503 Service Unavailable responses, showing the server is overloaded.

- When we filter traffic using Endpoints, it shows that Victim IP address is 192.168.1.10 and 50 other IP addresses trying to send packets to the victim within short period of time..



4. How many HTTP GET requests do you find?

Used HTTP/Requests statistics to check the number of HTTP requests. Total 2500 HTTP GET requests were found.



Wireshark - HTTP / Requests - file_lab2.pcap

Request Type	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
HTTP Requests by HTTP Host	2500				0.0255	100%	0.0200	49.000
bank.xfibility.com	2500				0.0255	100.00%	0.0200	49.000
/login	2500				0.0255	100.00%	0.0200	49.000

Display filter: Enter a display filter ...

Copy Save as... Close

5. What is the purpose of HTTP GET requests?

HTTP GET requests are used by the clients, especially web browsers, to request data from the webserver using HTTP/1.1 or HTTP/2. In this case using HTTP/1.1.

6. What is the significance of the 503-response code in the data?

It means the webserver is temporarily unable to handle a request. This happens due to overload with traffic or planned maintenance is conducting.

7. What is the IP address of the victim system? Show this using two different methods in Wireshark.

- We can use *Statics > Endpoints > IPv4* to filter all the IP addresses and see the victim IPv4 address 192.168.1.10

file_lab2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Wireshark - Endpoints - file_lab2.pcap

Endpoint Settings

Name resolution

Limit to display filter

Copy

Ethernet - 2	IPv4 - 51	IPv6	TCP - 2501	UDP										
Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization		
192.168.1.10	3,750	421 kB	1,250	164 kB	2,500	258 kB								
198.51.100.1	75	8 kB	50	5 kB	25	3 kB								
198.51.100.2	75	8 kB	50	5 kB	25	3 kB								
198.51.100.3	75	8 kB	50	5 kB	25	3 kB								
198.51.100.4	75	8 kB	50	5 kB	25	3 kB								
198.51.100.5	75	8 kB	50	5 kB	25	3 kB								
198.51.100.6	75	8 kB	50	5 kB	25	3 kB								
198.51.100.7	75	8 kB	50	5 kB	25	3 kB								

- We can use *http.response* display filter command to check the victim IP address. It sending the HTTP/1.1 503 Service Unavailable responses.

file_lab2.pcap

Filter: http.response

No.	Time	Source	Destination	Protocol	Length	Info
2	0.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
5	2.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
8	4.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
11	6.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
14	8.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
17	10.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
20	12.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
23	14.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
26	16.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
29	18.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
32	20.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
35	22.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
38	24.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
41	26.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
44	28.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
47	30.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
50	32.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
53	34.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
56	36.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
59	38.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
62	40.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
65	42.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable
68	44.100000	192.168.1.10	198.51.100.1	HTTP	131	[TCP ACKed unseen segment] HTTP/1.1 503 Service Unavailable

- How many unique IP addresses sent traffic to the victim? Using two different methods in Wireshark, show how you derived this.

- Endpoint statistics shows that Total 51 including victim, so **50** unique IPv4 addresses sent traffic to the victim.

file_lab2.pcap

File

Edit

View

Go

Capture

Analyze

Statistics

Telephony

Wireless

Tools

Help

Wireshark · Endpoints - file_lab2.pcap

Endpoint Settings

Name resolution

Limit to display filter

Copy

Map

Protocol

Bluetooth

BPv7

DCCP

✓

Ethernet

FC

FDDI

IEEE 802.11

IEEE 802.15.4

✓

IPv4

✓

IPv6

IPX

JXTA

Filter list for specific type

Ethernet · 2

IPv4 · 51

IPv6

TCP · 2501

UDP

Address

Packets

Bytes

Tx Packets

Tx Bytes

Rx Packets

Rx Bytes

Country

City

Latitude

Longitude

AS Number

AS Organization

192.168.1.10

3,750

421 kB

1,250

164 kB

2,500

258 kB

198.51.100.1

75

8 kB

50

5 kB

25

3 kB

198.51.100.2

75

8 kB

50

5 kB

25

3 kB

198.51.100.3

75

8 kB

50

5 kB

25

3 kB

198.51.100.4

75

8 kB

50

5 kB

25

3 kB

198.51.100.5

75

8 kB

50

5 kB

25

3 kB

198.51.100.6

75

8 kB

50

5 kB

25

3 kB

198.51.100.7

75

8 kB

50

5 kB

25

3 kB

198.51.100.8

75

8 kB

50

5 kB

25

3 kB

198.51.100.9

75

8 kB

50

5 kB

25

3 kB

198.51.100.10

75

8 kB

50

5 kB

25

3 kB

198.51.100.11

75

8 kB

50

5 kB

25

3 kB

198.51.100.12

75

8 kB

50

5 kB

25

3 kB

198.51.100.13

75

8 kB

50

5 kB

25

3 kB

198.51.100.14

75

8 kB

50

5 kB

25

3 kB

198.51.100.15

75

8 kB

50

5 kB

25

3 kB

198.51.100.16

75

8 kB

50

5 kB

25

3 kB

198.51.100.17

75

8 kB

50

5 kB

25

3 kB

198.51.100.18

75

8 kB

50

5 kB

25

3 kB

198.51.100.19

75

8 kB

50

5 kB

25

3 kB

198.51.100.20

75

8 kB

50

5 kB

25

3 kB

198.51.100.21

75

8 kB

50

5 kB

25

3 kB

198.51.100.22

75

8 kB

50

5 kB

25

3 kB

198.51.100.23

75

8 kB

50

5 kB

25

3 kB

198.51.100.24

75

8 kB

50

5 kB

25

3 kB

198.51.100.25

75

8 kB

50

5 kB

25

3 kB

- Conversations statistics showing that 50 IPv4 Addresses sent traffic to the Victim.

Conversation Settings		Ethernet · 1	IPv4 · 50	IPv6	TCP · 2500	UDP
Name resolution	Address A	Address B	Packets	Bytes	Stream ID	Rel Start
☐	198.51.100.1	192.168.1.10	75	8 kB	0	0.000000
☐	198.51.100.2	192.168.1.10	75	8 kB	1	1.000000
☐	198.51.100.3	192.168.1.10	75	8 kB	2	2.000000
☐	198.51.100.4	192.168.1.10	75	8 kB	3	3.000000
☐	198.51.100.5	192.168.1.10	75	8 kB	4	4.000000
☐	198.51.100.6	192.168.1.10	75	8 kB	5	5.000000
☐	198.51.100.7	192.168.1.10	75	8 kB	6	6.000000
☐	198.51.100.8	192.168.1.10	75	8 kB	7	7.000000
☐	198.51.100.9	192.168.1.10	75	8 kB	8	8.000000
☐	198.51.100.10	192.168.1.10	75	8 kB	9	9.000000
☐	198.51.100.11	192.168.1.10	75	8 kB	10	10.000000
☐	198.51.100.12	192.168.1.10	75	8 kB	11	11.000000
☐	198.51.100.13	192.168.1.10	75	8 kB	12	12.000000
☐	198.51.100.14	192.168.1.10	75	8 kB	13	13.000000
☐	198.51.100.15	192.168.1.10	75	8 kB	14	14.000000
☐	198.51.100.16	192.168.1.10	75	8 kB	15	15.000000
☐	198.51.100.17	192.168.1.10	75	8 kB	16	16.000000
☐	198.51.100.18	192.168.1.10	75	8 kB	17	17.000000
☐	198.51.100.19	192.168.1.10	75	8 kB	18	18.000000
☐	198.51.100.20	192.168.1.10	75	8 kB	19	19.000000
☐	198.51.100.21	192.168.1.10	75	8 kB	20	20.000000
☐	198.51.100.22	192.168.1.10	75	8 kB	21	21.000000
☐	198.51.100.23	192.168.1.10	75	8 kB	22	22.000000
☐	198.51.100.24	192.168.1.10	75	8 kB	23	23.000000
☐	198.51.100.25	192.168.1.10	75	8 kB	24	24.000000
☐	198.51.100.26	192.168.1.10	75	8 kB	25	25.000000

9. What HTTP method was used to flood the victim's system?

All the traffic use HTTP GET requests. We can confirm it using `http.request` display command, also you can use `http.request.method == "GET"` command also.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
3	1.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
4	2.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
6	3.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
7	4.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
9	5.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
10	6.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
12	7.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
13	8.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
15	9.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
16	10.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
18	11.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
19	12.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
21	13.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
22	14.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
24	15.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
25	16.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
27	17.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
28	18.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
30	19.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
31	20.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
33	21.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1
34	22.000000	198.51.100.1	192.168.1.10	HTTP	163	GET /login HTTP/1.1

10. What was the most common destination port used in this scenario?

Use Endpoint statistics to discover TCP protocol traffic and verified that all traffic was sent to TCP port 80 (HTTP).

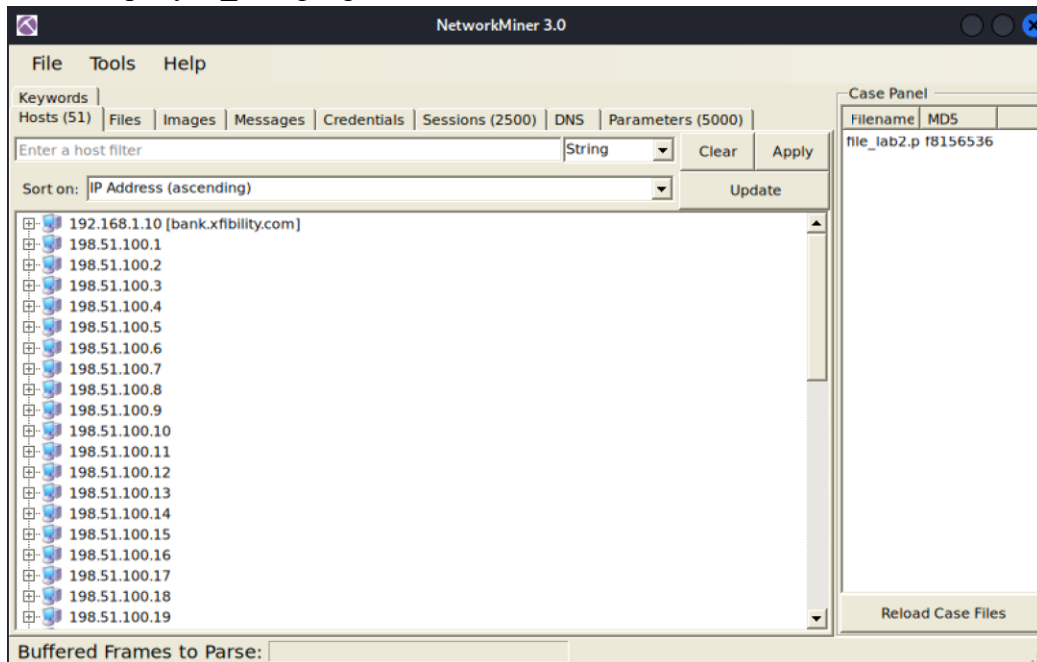
Endpoint Settings		Ethernet · 2	IPv4 · 51	IPv6	TCP · 2501	UDP
Name resolution	Address	Port	Packets	Bytes	Tx Packets	Tx Bytes
☐	192.168.1.10	80	3,750	421 kB	1,250	164 kB
☐	198.51.100.1	2022	2	234 bytes	1	103 bytes
☐	198.51.100.1	2274	2	234 bytes	1	103 bytes
☐	198.51.100.1	2446	1	103 bytes	1	103 bytes
☐	198.51.100.1	2672	1	103 bytes	1	103 bytes
☐	198.51.100.1	3664	1	103 bytes	1	103 bytes
☐	198.51.100.1	3896	2	234 bytes	1	103 bytes
☐	198.51.100.1	4135	2	234 bytes	1	103 bytes
☐	198.51.100.1	8159	2	234 bytes	1	103 bytes
☐	198.51.100.1	8628	1	103 bytes	1	103 bytes
☐	198.51.100.1	12254	1	103 bytes	1	103 bytes
☐	198.51.100.1	12335	1	103 bytes	1	103 bytes
☐	198.51.100.1	15026	2	234 bytes	1	103 bytes
☐	198.51.100.1	15087	2	234 bytes	1	103 bytes
☐	198.51.100.1	15327	2	234 bytes	1	103 bytes

Step -2

1. Opened the Network miner using `mono /opt/NetworkMiner_*/NetworkMiner.exe --noupdatecheck` command on Kali Linux Terminal.

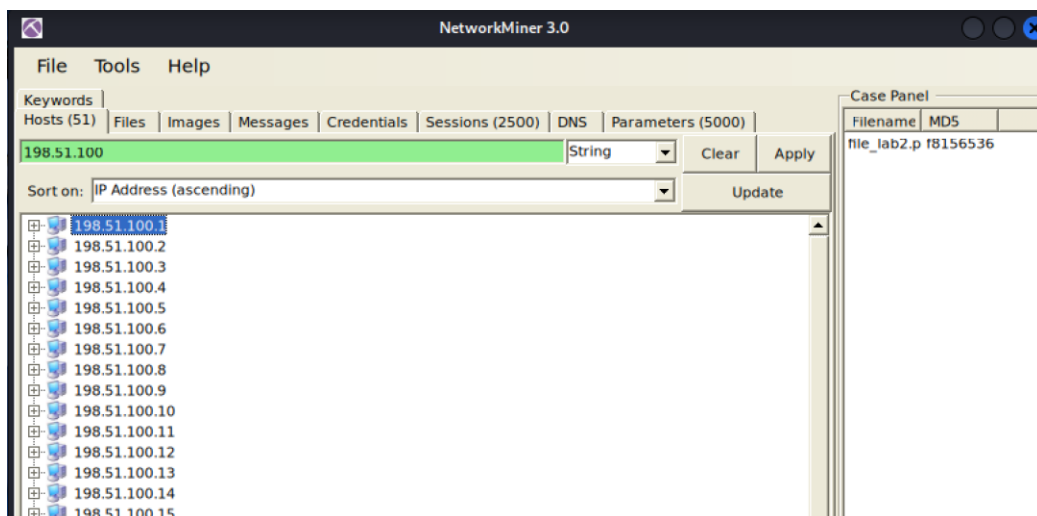
```
(kali@kali)-[~]  
$ mono /opt/NetworkMiner_*/NetworkMiner.exe --noupdatecheck  
Gtk not found (missing LD_LIBRARY_PATH to libgtk-x11-2.0.so.0?), using built-in colorscheme
```

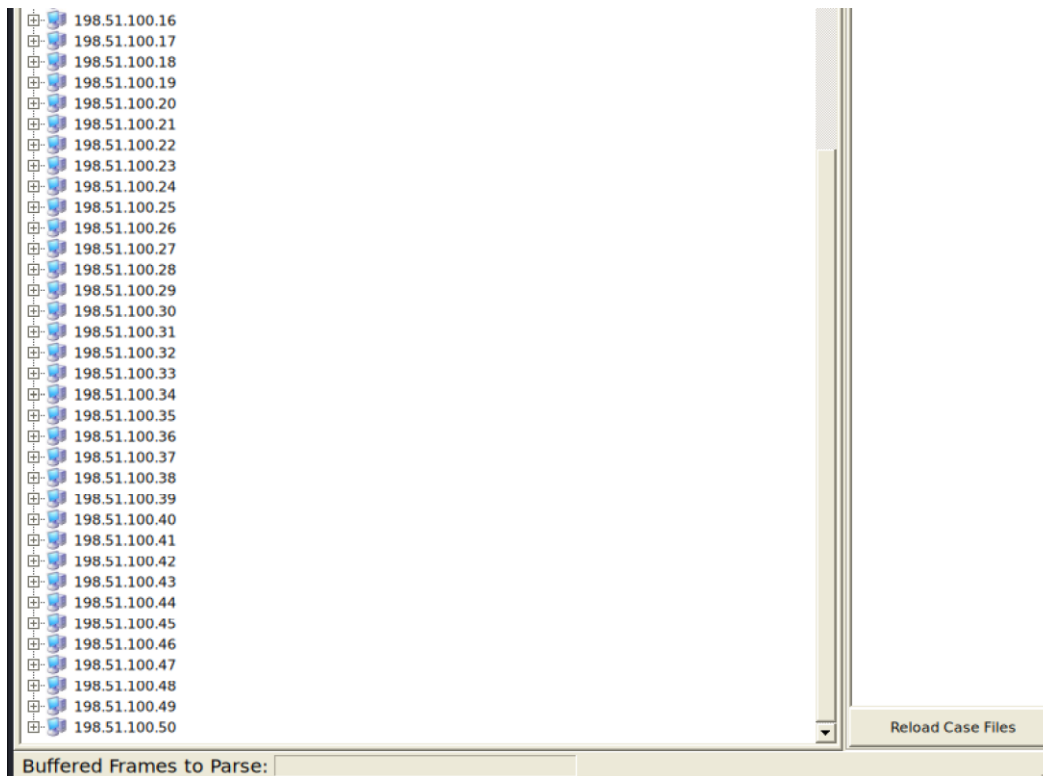
2. Open `file_lab2.pcap` file in Network Miner.



3. Identify relevant source IP address(es).

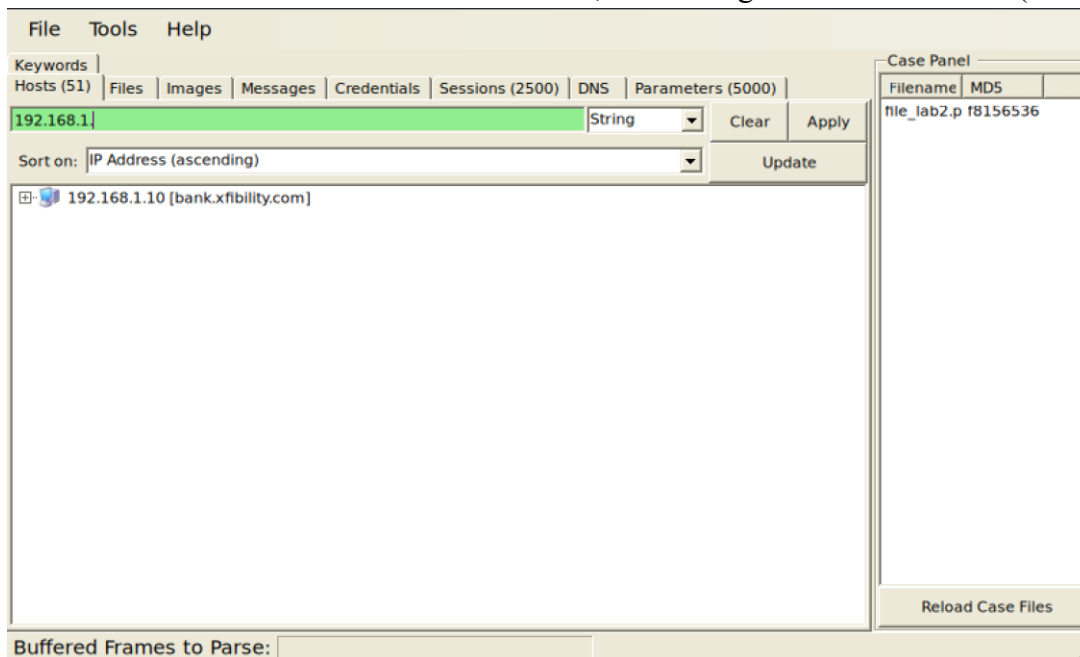
Source IP addresses range starting from 198.51.100.1 to 198.51.100.50, all together 50 IPv4 addresses.





4. Identify relevant destination IP address(es).

Destination IP address is 192.168.1.10, that belongs to bank web server(bank.xfibility.com)



Conclusion

This lab helps to cover the practical aspects of packet analysis using Wireshark and Network Miner. First, I did packet analysis using Wireshark and used different filters and statistics to understand the incident scenario. According to the bank security operation center, the firewall at the bank shows a massive spike in traffic to the webserver. Filter reports and statistics clearly show IP addresses of sources and destination, protocol and ports used for communication. It is clearly shown that this incident happened due to Distributed Denial of Service Attack. Statistics reports show, 50 IPV4 addresses ranging from 198.51.100.1 to 198.51.100.50 were trying to send HTTP -GET requests to the bank webserver 192.168.1.10.

The DDoS attacks mainly generate lots of false requests from different, or “Distributed,” sources, which drowns out legitimate requests (Chou & Groves, 2018). Same incident was shown in the bank webserver. The attacker uses a lot of sources IPv4 addresses to generate HTTP GET requests to access the webserver. The server responded repeatedly with HTTP/1.1 503 Service Unavailable response. The attacker targeted port 80 which belongs to HTTP protocol.

The bank must implement a detection system for this kind of traffic and DDoS mitigation services. The cyber threat landscape is complex and constantly changing, when a vulnerability identified, a tool that can exploit also developed within a short time of period (Calder, 2023). The security team can implement security controls by getting guidelines from National Institute of Standards and Technology (NIST) standards.

References

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3. Linode. (2023). *Understanding HTTP Requests: Structure, Methods & Examples*. Retrieved September 15, 2025, from <https://www.linode.com/docs/guides/http-get-request/>
4. Sanders, C. (2017). *Practical packet analysis* (3rd ed.).
5. Chou, E., & Groves, R. (2018). *Distributed denial of service (DDoS)*. O'Reilly Media, Inc.
6. Calder, A (2023). *Cyber Resilience*. IT Governance Publishing.